MSS 517

FLUID MECHANICS

BERNOULLI'S PRINCIPLE

Prepared by:
Angala, James Nicolle
Fulgencio, Michelle
Jayoma, Andrei
Magbanua, Arcelli
Pobocan, Angel Mae
Resuena, Genel Nicko
Siervo, Janella Rose

Preface

Welcome to Bernoulli's Principle Module!

Dear Learners,

This digital module is designed to help you in understanding the relationship between fluid speed and pressure. Inside, you will find clear explanations, real-life examples, and step-by-step problem-solving practice to guide you through the topic.

The activities in this module will not only master the equations and its derivations but also see how this principle applies the world around you.

Also, we would like to express our gratitude to everyone who helped this module possible. To our course professor and serve as one of the content validator **Mr. Joey Gaton,**thank you for your expertise, guidance, and valuable feedback which greatly improved this module. To **Ms. Tapan,** thank you for your support and encouragement throughout the development of this module. We also deeply thankful to **Mr. Renz and Mr. Norman,** for their review, constructive comments, and time ensuring this module is accurate and effective for learners.

Above all, we praise and glorify the Almighty God for the wisdom, strength, and patience to complete this module. We hope throughout this module, it helps make Bernoulli's Principle more understandable and meaningful for you.

Sincerely yours,
Angala, James Nicolle
Fulgencio, Michelle
Jayoma, Andrei
Magbanua, Arcelli
Pobocan, Angel Mae
Resuena, Genel Nicko
Siervo, Janella Rose
. - The Authors

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The Bernoulli's Principle



Key Terms

Bernoulli's Principle
An application of the conservation of mechanical energy to a moving fluid that describes the relationship between pressure, velocity, and height/elevation along a streamline.

Bernoulli's Equation
The mathematical formula used to calculate fluid dynamics, expressed as

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

Pressure (P)
The force exerted by a fluid per unit area, expressed in Pascals (Pa).

Velocity (v)
The speed at which the fluid flows, measured in meters per second (m/s).

Fluid Density (ρ)
The mass per unit volume of a fluid, quantified in kilograms per cubic meter (kg/m^3).

Height / Elevation (h)
The vertical position of the fluid relative to a reference level, measured in meters (m).

Acceleration Due to Gravity (g)
A constant acceleration value representing gravity, standardly set at 9.8 m/s^2.

Streamline
The continuous path followed by individual fluid particles as they move.

Steady, Incompressible Flow
A smooth type of fluid flow where density remains constant and fluid properties do not change over time, assuming negligible frictional losses.

The Bernoulli's Principle



Learning Objectives

- Explain the relationship among pressure, velocity, density, and elevation in fluid flow using Bernoulli's Equation;
- Apply Bernoulli's Equation to solve fluid-flow problems involving pressure, velocity, density, and elevation using appropriate problem-solving procedures and units; and
- Demonstrate appreciation of the importance of Bernoulli's Principle by identifying at least two of its applications in water-supply systems and other engineering contexts.



Looking Back

Directions: Read the question carefully and write the letter of the correct answer.

1. What principle explains the conservation of mass in fluid flow?

A. Pascal's Principle
B. Bernoulli's Principle
C. Continuity Principle
D. Archimedes Principle

2. Which of the following represents the continuity equation for an incompressible fluid?

A. $F= ma$
B. $d= vt$
C. $A_1v_1 = A_2v_2$
D. $P_1+ pgh = constant$

3. The fluid continuously flows through a cross-sectional pipe with an area of 0.04 and a velocity of 2 m/s. The pipe narrows to 0.02 m^2. What do you think will happen to the velocity of the water in the narrower area?

A. 2 m/s
B. 4 m/s
C. 6 m/s
D. 8 m/s

4. The students are tasked to water the plants in the school garden, they use a hose to quickly water the plants, the opening can be adjusted. The students adjusted the nozzle opening to smaller while the same amount of water continues to flow, what do you think will likely happen to the velocity?

A. The water will stop flowing.
B. The water will become dense.
C. The water velocity will increase.
D. The water velocity will decrease.

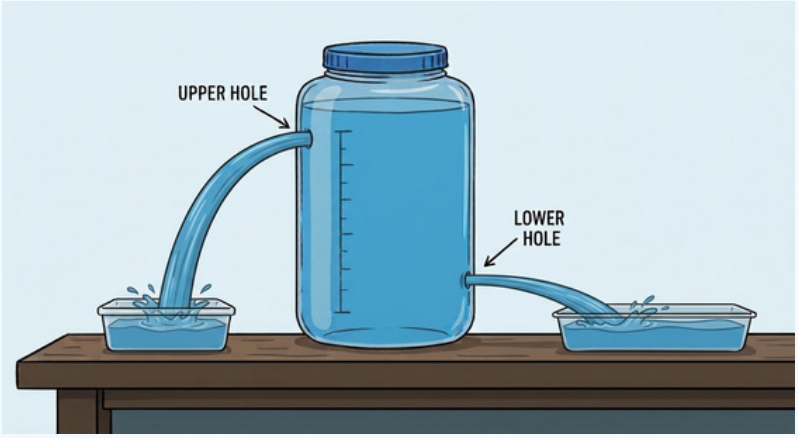
5. A student claims, “When the pipe becomes narrower, the velocity of the water decreases because there is less space for the water to move.” Based on the continuity equation, how should this claim be explained?

A. Correct, because the water will stop moving in narrow pipes.
B. Correct, because the less area always means the velocity is also less.
C. Incorrect, because the velocity is independent of the cross-sectional area.
D. Incorrect, because velocity increases when area decreases, assuming constant flow rate.



What do you think?

Directions: Read the scenario carefully and encircle the letter of the best answer. Provide your explanation on the space provided below.



Two holes are made in a water container. One hole is located near the top, while the other is located closer to the bottom. Which hole will produce a water jet that travels farther?

A. The hole near the top
B. The hole near the bottom
C. Both will travel the same distance

Question: Why did you think that result will most likely happen? Explain your prediction in 4-5 sentences.

_____
_____
_____
_____
_____
_____
_____

Your prediction is connected to the relationship between pressure, velocity, and elevation. In this module, you will discover how Bernoulli's Principle explains this phenomenon.

Think it Through!

Do you wear perfumes? Have you ever wondered how the nozzle of a perfume bottle can atomize the perfume when sprayed?



Figure 1. Perfume Bottle

A perfume bottle is consist of a container, tube, pump, and a nozzle. When you spray a perfume, the bulb that you press squeezes out the air that is in the tube inside the container through compression. The release of this air creates lower pressure on the nozzle. On the other hand, the pressure in the perfume reservoir remains constant because it is not directly affected by the airflow. That is why, a pressure difference between the nozzle and perfume reservoir is created.

This pressure difference results to the forced movement of the perfume out of the bottle. It is due to the reason that high pressure areas move to the areas that has lower pressure. The perfume comes out through the smaller tubes and gets mixed with the air that has high velocity, producing finer mist.

The principle behind this mechanism is called **Bernoulli's Principle**.

Now, Let's Make Sense of It

Bernoulli's principle is an application of the conservation of mechanical energy to a moving fluid. It explains the relationship between **pressure, velocity, and height** as a fluid flows along a streamline.

Bernoulli's Equation is written as: $P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$

where (**P**) is pressure, (**ρ**) is fluid density, (**v**) is fluid velocity, (**g**) is acceleration due to gravity, and (**h**) is height.

The equation shows that changes in pressure, velocity, and height are related. For example, when water flows through a pipe that becomes narrower, its velocity increases and pressure decreases because the same amount of water must pass through a smaller cross-sectional area. If the height remains the same, the increase in velocity is accompanied by a decrease in pressure.

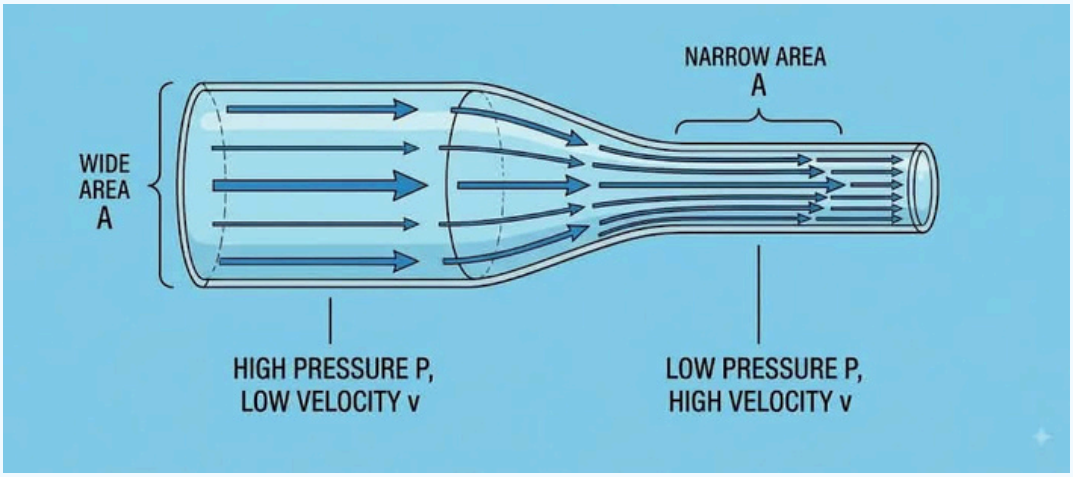


Figure 2. Water Flows in Pipe with Varying Cross-sectional Areas

Bernoulli's Equation is mainly used for **steady, incompressible flow along a streamline with negligible frictional losses**. It may not be suitable when there are significant frictional losses, highly turbulent flow, or compressible fluids.

Now, Let's Make Sense of It

But what will happen if the fluid flows from a different elevation (height)? Does pressure and velocity stay the same? Or the concept will alter?

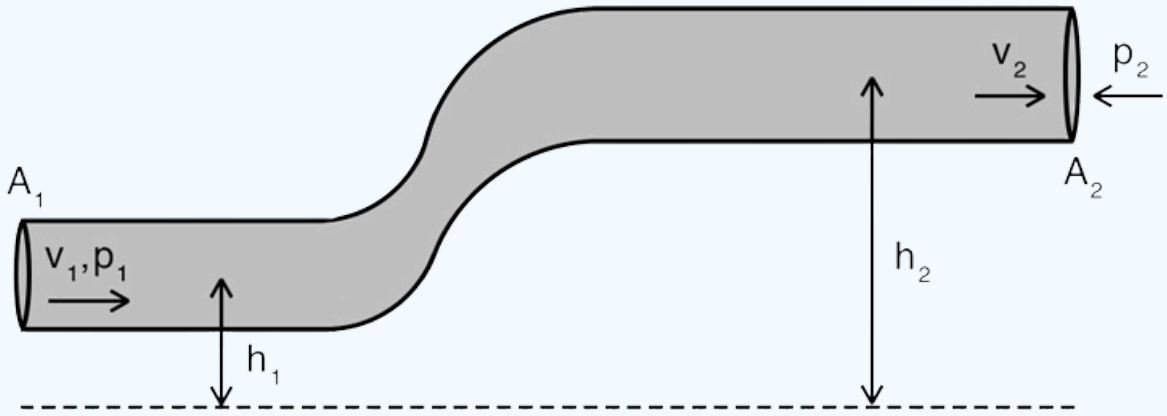
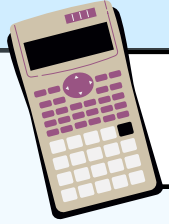


Figure 3. Water Flows in Pipe with Varying Cross-sectional Areas and Elevation
https://share.google/2F7gQi7J9DxO0xd5b

When a fluid flows from a lower elevation to a higher elevation, its potential energy increases because of gravity. As a result, additional energy may be needed to move the fluid upward, such as energy supplied by a pump. Bernoulli's Principle can then be used to analyze how the fluid's pressure, velocity, and height are related.

The size of the pipe also affects the fluid's velocity and pressure. When the fluid flows through a wider pipe, its velocity is generally lower. When it enters a narrower pipe, its velocity increases. If the height remains the same, this increase in velocity is accompanied by a decrease in pressure.

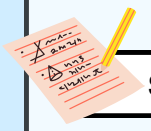
On the other hand, when a fluid flows from a higher elevation to a lower elevation, its potential energy decreases. If its velocity remains constant, the pressure may increase as the fluid moves downward. This shows how changes in elevation can affect the pressure and velocity of a moving fluid.



Formula $$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

Key Variables	Symbol	SI Unit	Constant
Pressure	P	Pa	-
Density (Water)	ρ	kg/m³	1000 kg/m³
Velocity	v	m/s	-
Acceleration due to gravity	g	m/s²	9.8 m/s²
Height	h	m	-

Table 1. Summary of Variables, Symbols, and Constants in Bernoulli's Equation



Sample Problem 1

During a physics laboratory activity, second-year students at the University of Caloocan City investigate fluid dynamics by analyzing water flowing steadily through a horizontal pipe. The speed of the water at section A is 1.2 m/s with a pressure of 240 kPa, while the speed at section B is 4.8 m/s. Assume the density of water is 1000 kg/m³. What is the fluid pressure at section B?

Given
(Write down all the facts, numbers, and values provided in the problem.)

$v_1 = 1.2$ m/s
$v_2 = 4.8$ m/s
$P_1 = 240$ kPa $= 240{,}000$ Pa
$\rho = 1000$ kg/m³
Horizontal pipe $\rightarrow h_1 = h_2$

Required
(State the unknown value you need to find.)

P_2

Equation
(Write the correct formula needed to solve the problem.)

$$P_1 + \frac{1}{2}\rho v_1^2 + \cancel{\rho g h_1} = P_2 + \frac{1}{2}\rho v_2^2 + \cancel{\rho g h_2}$$ (neglecting the gravity since horizontal pipe and the fluid didn't elevate nor fall)

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$ (transpose $\frac{1}{2}\rho v_2^2$ to the other side)

$$P_1 + \frac{1}{2}\rho v_1^2 - \frac{1}{2}\rho v_2^2 = P_2$$ (factor out common values)

$$P_1 + \frac{1}{2}\rho\left(v_1^2 - v_2^2\right) = P_2$$ (rewrite with target variable first)

$$P_2 = P_1 + \frac{1}{2}\rho\left(v_1^2 - v_2^2\right)$$

Solution
(Substitute the given into the formula and do the step-by-step.)

$$P_2 = 240{,}000\ Pa + \frac{1}{2}(1000\ kg/m^3)(1.2^2\ m/s - 4.8^2\ m/s)$$ (start by computing the exponent)

$$P_2 = 240{,}000\ Pa + \frac{1}{2}(1000\ kg/m^3)(1.44\ m/s - 23.04\ m/s)$$ (subtract the value inside the parentheses)

$$P_2 = 240{,}000\ Pa + \frac{1}{2}(1000\ kg/m^3)(-21.6\ m/s)$$ (mutiply $\frac{1}{2}(1000\ kg/m^3)(-21.6\ m/s)$)

$$P_2 = 240{,}000\ Pa + (-10{,}800)$$ (compute)

$$P_2 = 229{,}200\ Pa$$ (convert the answer to kilopascals by dividing by 1,000)

$$P_2 = 229{,}200\ Pa \div 1000$$ (divide the total sum by 1000 to convert the pressure into kilopascals)

Answer
(State your final computed answer clearly, including the correct units.)

$$P_2 = 229.2\ kPa$$

Sample Problem 2

While checking the household water system, you notice water flowing from a wide main supply pipe located at an elevation of 0.5 m above the ground into a narrower overhead pipe leading to the shower, which is located at an elevation of 2.5 m above the ground. The speed of the water at the main pipe is 1.5 m/s, and the water accelerates to a speed of 5.5 m/s as it reaches the smaller pipe with a pressure of 180 kPa. Assuming the density of water is 1000 kg/m³, what is the fluid pressure at the main supply pipe?

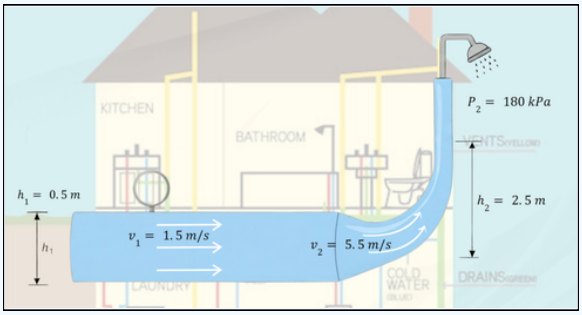


Figure 4. Water Flows in Pipe with Varying Cross-sectional Areas and Elevation
https://www.gharpedia.com/blog/residential-plumbing-system/

Given
(Write down all the facts, numbers, and values provided in the problem.)

$v_1 = 1.5$ m/s
$v_2 = 5.5$ m/s
$P_2 = 180$ kPa $= 180{,}000$ Pa
$\rho = 1000$ kg/m³
$h_1 = 0.5$ m
$h_2 = 2.5$ m

Required
(State the unknown value you need to find.)

$$P_1$$

Equation
(Write the correct formula needed to solve the problem.)

$$P_1 + \boxed{\tfrac{1}{2}\rho v_1^2 + \rho g h_1} = P_2 + \tfrac{1}{2}\rho v_2^2 + \rho g h_2$$

isolate P_2 by moving $\frac{1}{2}\rho v_1^2$ and $\rho g h_1$ to the right side

$$P_1 = P_2 + \left(\tfrac{1}{2}\rho v_2^2 - \tfrac{1}{2}\rho v_1^2\right) + \left(\rho g h_2 - \rho g h_1\right)$$

factor out the common terms

$$P_1 = P_2 + \tfrac{1}{2}\rho\left(v_2^2 - v_1^2\right) + \rho g\left(h_2 - h_1\right)$$

Solution
(Substitute the given into the formula and do the step-by-step.)

$$P_1 = 180{,}00\ Pa + \left[\tfrac{1}{2}\left(1000\ kg/m^3\right)\left((5.5 m/s)^2 - (1.5\ m/s)^2\right)\right] + \left[\left(1000\ kg/m^3\right)\left(9.8\ m/s^2\right)(2.5m - 0.5\ m)\right]$$

(calculate the exponents by squaring the velocities)

$$P_1 = 180{,}00\ Pa + \left[\tfrac{1}{2}\left(1000\ kg/m^3\right)(30.25\ m/s - 2.25\ m/s)\right] + \left[\left(1000\ kg/m^3\right)\left(9.8\ m/s^2\right)(2.5m - 0.5\ m)\right]$$

(subtract the values inside the parentheses)

$$P_1 = 180{,}00\ Pa + \left[\tfrac{1}{2}\left(1000\ kg/m^3\right)(28\ m/s)\right] + \left[\left(1000\ kg/m^3\right)\left(9.8\ m/s^2\right)(2\ m)\right]$$

$$P_1 = 180{,}00\ Pa + 14{,}000 + 19{,}600$$

(multiply the terms within their respective brackets and add the resulting values together with the initial pressure)

$$P_1 = 213{,}600 \div 1000$$

(divide the total sum by 1000 to convert the pressure into kilopascals)

Answer
(State your final computed answer clearly, including the correct units.)

$$P_1 = 213.6\ kPa$$

Sample Problem 3

When examining a multi-story home's plumbing system during renovation, you measure the water flowing through a horizontal main water supply line. The fluid pressure at the main pipe Section 1) is 240 kPa with an initial water speed of 1.2 m/s. As the water moves into a narrower section, the fluid pressure drops to 180 kPa. Assuming the density of water is 1000 kg/m³ and the system remains at the same height level, what is the speed of the water at Section 2?

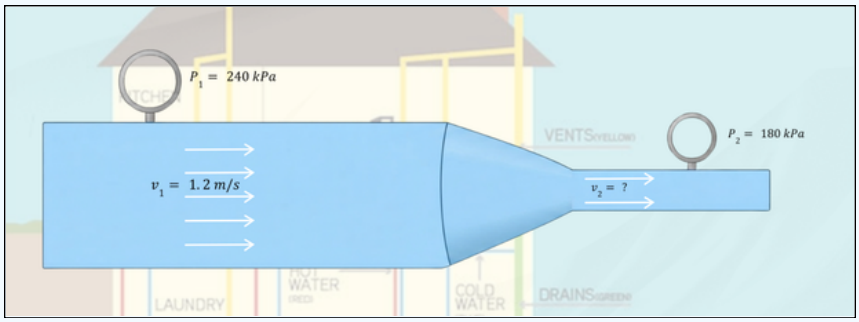


Figure 5. Water Flow in a Horizontal Pipe with Varying Cross-Sectional Area
https://www.gharpedia.com/blog/residential-plumbing-system/

Given
(Write down all the facts, numbers, and values provided in the problem.)

$v_1 = 1.2$ m/s
$P = 240$ kPa $= 240{,}000$ Pa
$P_2 = 180$ kPa $= 180{,}000$ Pa
$\rho = 1000$ kg/m³
Horizontal pipe → $h_1 = h_2$

Required
(State the unknown value you need to find.)

$$v_2$$

Equation
(Write the correct formula needed to solve the problem.)

$$P_1 + \frac{1}{2}\rho v_1^2 + \cancel{\rho g h_1} = P_2 + \frac{1}{2}\rho v_2^2 + \cancel{\rho g h_2}$$

(neglecting the gravity since horizontal pipe and the fluid didn't elevate nor fall)

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

(transpose P_2 to the other side)

$$P_1 + \frac{1}{2}p v_1^2 - P_2 = \frac{1}{2}p v_2^2$$

$$\frac{1}{2}p v_2^2 = (P_1 - P_2) + \frac{1}{2}p v_1^2$$

(rearrange to isolate the v_2^2 term)

$$2\left[\cancel{\frac{1}{2}}p v_2^2 = (P_1 - P_2) + \cancel{\frac{1}{2}}p v_1^2\right]2$$

(multiply both side by 2 to clear the fraction)

$$\frac{\cancel{p}v_2^2}{\cancel{p}} = \frac{2(P_1 - P_2)}{p} + \frac{\cancel{p}v_1^2}{\cancel{p}}$$

(devide both side by ρ)

$$\sqrt{v_2^2} = \sqrt{\frac{2(P_1 - P_2)}{P} + v_1^2}$$

(take the square root on both sides to eliminate the exponent of v_2^2)

$$v_2 = \sqrt{\frac{2(P_1 - P_2)}{P} + v_1^2}$$

Solution
(Substitute the given into the formula and do the step-by-step.)

$$v_2 = \sqrt{\frac{2(P_1 - P_2)}{p} + v_1^2}$$

$$v^2 = \sqrt{\frac{2\,(240{,}00\ Pa - 180{,}000\ Pa)}{1000\ kg/m^3} + (1.2\ m/s)^2}$$

(Subtract the values inside the parentheses first)

Sample Problem 3

When examining a multi-story home's plumbing system during renovation, you measure the water flowing through a horizontal main water supply line. The fluid pressure at the main pipe Section 1) is 240 kPa with an initial water speed of 1.2 m/s. As the water moves into a narrower section, the fluid pressure drops to 180 kPa. Assuming the density of water is 1000 kg/m³ and the system remains at the same height level, what is the speed of the water at Section 2?

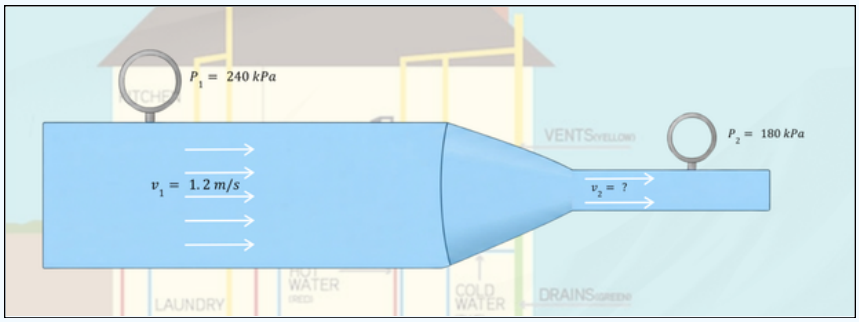
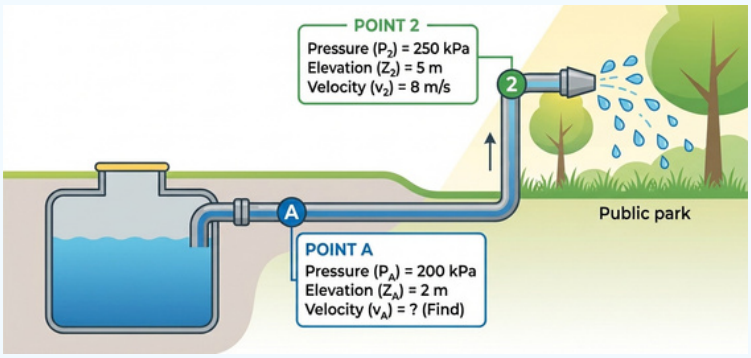


Figure 5. Water Flow in a Horizontal Pipe with Varying Cross-Sectional Area
https://www.gharpedia.com/blog/residential-plumbing-system/

Solution (Substitute the given into the formula and do the step-by-step.)	$v^2 = \sqrt{\frac{2\,(60{,}000\ Pa)}{1000\ kg/m^3} + (1.2\ m/s)^2}$ multiply 2 (60,000 Pa) $v_2 = \sqrt{\frac{120{,}000\ Pa}{1000\ kg/m^3} + (1.2\ m/s)^2}$ divide 120,000 Pa to 1000 kg/m³ $v_2 = \sqrt{120 + (1.2\ m/s)^2}$ square the initial velocity $v_2 = \sqrt{120 + 1.44\ m/s}$ add the values $v_2 = \sqrt{121.44}$ m/s get the square root
Answer (State your final computed answer clearly, including the correct units.)	$v_2 = 11.02\ m/s$

Sample Problem 4

A water sprinkler system at a public park in Quezon City uses a pipeline to transport water from a storage reservoir to a sprinkler nozzle. The water has a pressure of 200 kPa at Point A and an elevation of 2 m. While at Point 2, the water has a pressure of 250 kPa with a velocity of 8 m/s and an elevation of 5 m. Assume the water density is 1,000 kg/m³, gravitational acceleration is 9.8 m/s², what is the velocity of water at point A?



Given
(Write down all the facts, numbers, and values provided in the problem.)

v_2 = 8 m/s
P_1 = 200 kPa = 200 000 Pa
P_2 = 250 kPa = 250 000 Pa
h_1 = 2m
h_2 = 5m
g = 9.8 m/s^2
ρ = 1000 kg/m³

Required
(State the exact question or the unknown value you need to find.)

v_1 = ?

Equation
(Write the correct formula needed to solve the problem.)

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$$\frac{1}{2}\rho v_1^2 = (P_2 - P_1) + \frac{1}{2}\rho v_2^2 + \rho g(h_2 - h_1)$$

$$2\left[\frac{1}{2}\rho v_1^2 = (P_2 - P_1) + \frac{1}{2}\rho v_2^2 + \rho g(h_2 - h_1)\right]$$

$$\rho v_1^2 = 2(P_2 - P_1) + \rho v_2^2 + 2\rho g(h_2 - h_1)$$

$$\frac{\rho v_1^2}{\rho} = \frac{2(P_2 - P_1)}{\rho} + \frac{\rho v_2^2}{\rho} + \frac{2\rho g(h_2 - h_1)}{\rho}$$

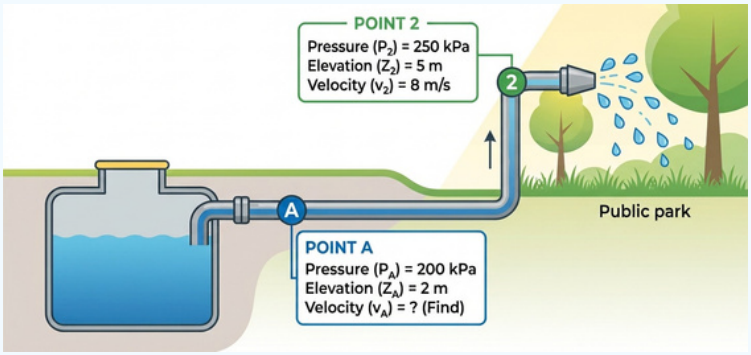
$$\frac{\rho v_1^2}{\rho} = \frac{2(P_2 - P_1)}{\rho} + \frac{\rho v_2^2}{\rho} + \frac{2\rho g(h_2 - h_1)}{\rho}$$

$$\sqrt{v_1^2} = \sqrt{\frac{2(P_2 - P_1)}{\rho} + v_2^2 + 2g(h_2 - h_1)}$$

$$v_1 = \sqrt{\frac{2(P_2 - P_1)}{\rho} + v_2^2 + 2g(h_2 - h_1)}$$

Sample Problem 4

A water sprinkler system at a public park in Quezon City uses a pipeline to transport water from a storage reservoir to a sprinkler nozzle. The water has a pressure of 200 kPa at Point A and an elevation of 2 m. While at Point 2, the water has a pressure of 250 kPa with a velocity of 8 m/s and an elevation of 5 m. Assume the water density is 1,000 kg/m³, gravitational acceleration is 9.8 m/s², what is the velocity of water at point A?



Solution

(Substitute the given into the formula and do the step-by-step.)

$$v_1 = \sqrt{\frac{2(250\,000\ Pa - 200\,000\ Pa)}{1000\ kg/m^3} + \left(8\frac{m}{s}\right)^2 + \left[2\left(9.8\frac{m}{s^2}\right)\right](5m - 2m)}$$

$$v_1 = \sqrt{\frac{2(50\,000\ Pa)}{1000\ kg/m^3} + 64\ \frac{m^2}{s^2} + (19.6\frac{m}{s^2})(3m)}$$

$$v_1 = \sqrt{\frac{100\,000\ Pa}{1000\ kg/m^3} + 64\ \frac{m^2}{s^2} + 58.8\ \frac{m^2}{s^2}}$$

$$v_1 = \sqrt{100\frac{m^2}{s^2} + 64\ \frac{m^2}{s^2} + 58.8\ \frac{m^2}{s^2}}$$

$$v_1 = \sqrt{222.8\ \frac{m^2}{s^2}}$$

$$v_1 = 14.93\ m/s$$

Answer

(State your final computed answer clearly, including the correct units.)

$$v_1 = 14.93\ m/s$$

Sample Problem 5

A water storage tank in a residential area in Caloocan City supplies water through a pipe to a lower-level outlet. At point A, the water has a pressure of 180 kPa and a velocity of 2 m/s. The elevation at this specific point is unknown. While at Point B, the water has a pressure of 250 kPa that makes the velocity of water 5 m/s. This has an elevation of 2 m above the ground. Assume that the water density is 1,000 kg/m³ and the gravitational acceleration is 9.8 m/s², what is the height at point A?



Given
(Write down all the facts, numbers, and values provided in the problem.)

$v_1 = 2$ m/s
$v_2 = 5$ m/s
$P_1 = 180$ kPa = 180 000 Pa
$P_2 = 250$ kPa = 250 000 Pa
$h_2 = 2$m
$\rho = 1000$ kg/m³
g = 9.8 m/s^2

Required
(State the exact question or the unknown value you need to find.)

$h_1 = ?$

Equation
(Write the correct formula needed to solve the problem.)

$$\boxed{P_1 + \frac{1}{2}\rho v_1^2} + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

(Rearrange the original formula to isolate the term with the required variable through transposition.)

$$\rho g h_1 = (P_2 - P_1) + \frac{1}{2}\rho(v_2^2 - v_1^2) + \rho g h_2$$

$$\frac{\rho g h_1}{\rho g} = \frac{(P_2 - P_1)}{\rho g} + \frac{\rho(v_2^2 - v_1^2)}{2\rho g} + \frac{\rho g h_2}{\rho g}$$

(Divide both sides by ρg.)

$$\frac{\cancel{\rho g} h_1}{\cancel{\rho g}} = \frac{(P_2 - P_1)}{\rho g} + \frac{\cancel{\rho}(v_2^2 - v_1^2)}{2\cancel{\rho} g} + \frac{\cancel{\rho g} h_2}{\cancel{\rho g}}$$

$$h_1 = \frac{(P_2 - P_1)}{\rho g} + \frac{(v_2^2 - v_1^2)}{2g} + h_2$$

Solution
(Substitute the given into the formula and do the step-by-step.)

$$h_1 = \frac{(250\,000\,Pa - 180\,000\,Pa)}{(1000 kg/m^3)(9.8 m/s^2)} + \frac{(5\,m/s)^2 - (2\,m/s)^2}{2\,(9.8 m/s^2)} + 2m$$

$$h_1 = \frac{(70000\,Pa)}{(9800 kg/m^2 \cdot s)} + \frac{21 m^2/s^2}{19.6\,m/s^2} + 2m$$

$$h_1 = (7.1428571429m) + 1.0714285714m + 2m$$

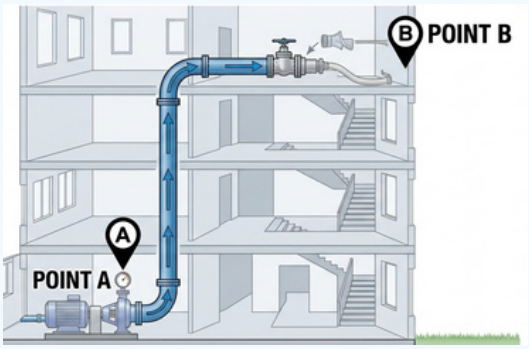
$$h_1 \text{ - } 10.21\,m$$

Answer
(State your final computed answer clearly, including the correct units.)

$$h_1 = 10.21\,m$$

Sample Problem 6

The water pipe used in a fire protection system to deliver water from a pump on the ground floor to a fire hose connection on an upper floor. The water has a pressure of 350 kPa at Point A and a velocity of 4 m/s with an elevation of 1.5 m above the ground. At Point B, the water has a pressure of 250 kPa and a velocity of 6 m/s. Assume the water density is 1,000 kg/m³, gravitational acceleration is 9.8 m/s², what is the elevation at point B?



Given

(Write down all the facts, numbers, and values provided in the problem.)

$v_1 = 4$ m/s
$v_2 = 6$ m/s
$P_1 = 350$ kPa = 350 000 Pa
$P_2 = 250$ kPa = 250 000 Pa
$h_1 = 1.5$m
$\rho = 1000$ kg/m³
g = 9.8 m/s^2

Required

(State the exact question or the unknown value you need to find.)

$h_2 = ?$

Equation

(Write the correct formula needed to solve the problem.)

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$$\rho g h_2 = (P_1 - P_2) + \frac{1}{2}\rho(v_1^2 - v_2^2) + \rho g h_1$$

$$\frac{\rho g h_2}{\rho g} = \frac{(P_1 - P_2)}{\rho g} + \frac{\rho(v_1^2 - v_2^2)}{2\rho g} + \frac{\rho g h_1}{\rho g}$$

$$\frac{\rho g h_2}{\rho g} = \frac{(P_1 - P_2)}{\rho g} + \frac{\rho(v_1^2 - v_2^2)}{2\rho g} + \frac{\rho g h_1}{\rho g}$$

$$h_2 = \frac{(P_1 - P_2)}{\rho g} + \frac{(v_1^2 - v_2^2)}{2g} + h_1$$

Solution

(Substitute the given into the formula and do the step-by-step.)

$$h_2 = \frac{(350\,000\ Pa - 250\,000\ Pa)}{(1000 kg/m^3)(9.8 m/s^2)} + \frac{(4m/s)^2 - (6m/s)^2}{2(9.8 m/s^2)} + 1.5m$$

$$h_2 = \frac{(100\,000\ Pa)}{(9800 kg/m^2 \cdot s)} + \frac{(16 m^2/s^2) - (36 m^2/s^2)}{19.6\ m/s^2} + 1.5m$$

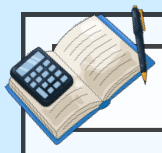
$$h_2 = \frac{(100\,000\ Pa)}{(9800 kg/m^2 \cdot s)} + \frac{(-20\ m^2/s^2)}{19.6\ m/s^2} + 1.5m$$

$$h_2 = 10.2040816327m + (-1.0204081633\ m) + 1.5m$$

Answer

(State your final computed answer clearly, including the correct units.)

$$h_2 = 10.68\ m$$



Follow the Flow

Objective: Demonstrate appreciation of the importance of Bernoulli's Principle by identifying its applications in real-life fluid-flow systems.

Directions: Read the situation carefully. Answer the question in 5–7 sentences. Use Bernoulli's Equation and appropriate scientific concepts to support your explanation.

Question: Water flows through a pipeline from a lower section to a higher section. As the water moves upward, the pipe also becomes narrower. Using Bernoulli's Equation, explain how the changes in velocity and elevation may affect the pressure of the water. In your explanation, discuss the role of fluid density and how the four quantities: pressure, velocity, density, and elevation, are related under ideal-flow conditions.



Try This!

Objective: Demonstrate appreciation of the importance of Bernoulli's Principle by identifying its applications in real-life fluid-flow systems.

Directions: Read the situation carefully. Answer the question in 5–7 sentences. Use Bernoulli's Equation and appropriate scientific concepts to support your explanation.

1. A doctor is administering IV fluid to a patient. The IV fluid flows through a section of tubing that becomes narrower. At Point 1, the fluid has a pressure of 120 kPa, a velocity of 3.0 m/s, and an elevation of 1.0 cm. At Point 2, the tubing is narrower, causing the fluid velocity to increase to 6.0 m/s. Point 2 is at an elevation of 0.5 cm. If the density of the IV fluid is approximately 1000 kg/m^3 and g=9.8m/s^2, what is the pressure at Point 2?

2. Students are conducting an experiment using a water pipe system in their school laboratory. At Point 1, the water has a pressure of 200 kPa, a velocity of 2.0 m/s, and is 2.0 m above the ground. At Point 2, the pressure is 150 kPa, and the pipe is located 5.0 m above the ground. Assume the density of water is 1,000 kg/m^3 and gravitational acceleration is 9.81 m/s^2, determine the velocity of water at Point 2.

3. A machine uses a horizontal pipe to transport oil from Point A to Point B. At Point A, the oil flows with a velocity of 3.0 m/s, while its pressure is unknown. At Point B, the oil has a velocity of 7.0 m/s and a pressure of 200 kPa. The density of the oil is 850 kg/m³. Assume there is no significant energy loss throughout the flow.

4. The BSE-SCIENCE students investigate a tank that supplies water to a household for their laboratory. They found out that at point A, the water has a pressure of 250 kPa, an elevation of 2 m, and an unknown velocity. At point B, the water has a pressure of 230 kPa, a velocity of 4 m/s, and an elevation of 5 m. The density of water is 1000 kg/m³. Assuming there is no energy loss due to friction, what is the velocity of the water at Point A?

5. A school sink supplies water using an elevated pipe connecting Point A to Point B. At point A, the water has a pressure of 280 kPa, velocity of 2.5 m/s, and is elevated to 1m. At the point B, the pressure decreases to 210 kPa, while the velocity increases to 5 m/s. Assume the water has a density of 1000 kg/m³ and no energy loss due to friction. What is the elevation of point B?

Where Do We See Bernoulli's Principle?

Objective: Demonstrate appreciation of the importance of Bernoulli's Principle by identifying its applications in real-life fluid-flow systems.

Directions: Think of two real-life systems or situations that involve the movement of fluids, such as water-supply systems, household plumbing, pumps, pipes, or other engineering applications.

For each example, complete the following:

Real-life Application	Where is Bernoulli's Principle involved?	Why is it important?
1.		
2.		

Reference

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